25901403 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMBG201 - COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Explain the purpose of nucleotide sequence databases. [U] / [CO1]
2. Define the term "Knowledge Discovery in Databases (KDD). [U] / [CO2]
3. What was the main goal behind developing Python? [K] / [CO3]
4. What is flexible docking? [K] / [CO4]
5. Name any one ligand database. [K] / [CO4]

[1x5 = 5]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

6. Describe agglomerative clustering. [K] / [CO2]

7. What is meant by portability in Python? [U] / [CO3]

8. Explain the concept of Information Gain in the context of tree induction. [U] / [CO2]

9. Distinguish between native and denatured proteins. [U] / [CO1]

10. State two applications of XML. [U] / [CO2]

11. State any two features of Python strings. [K] / [CO3]

12. What is virtual screening? [K] / [CO4]

13. What is a biological database? [K] / [CO1]

14. What is meant by database integration in PHP? [U] / [CO2]

15. What is the importance of indentation in Python files? [K] / [CO3]

16. Define binding site prediction. [K] / [CO4]

17. What is meant by gene annotation? [K] / [CO1]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

18. Describe how secondary databases help researchers. [U] / [CO1]

19. Explain how primary sequence of aminoacid contribute to the 3D function of a protein. [U] / [CO1]

20. Explain the role of web log files in web usage mining. [K] / [CO2]

21. Explain the role of KEGG in pathway analysis and functional genomics. [U] / [CO1]

22. Describe hierarchical clustering and its application in gene expression analysis. [U] / [CO1]

23. Describe the key assumptions behind the Naive Bayes classifier. [U] / [CO2]

24. Explain structure-based and ligand-based drug design approaches with suitable examples. [K] / [CO4]

[6x5 = 30]

Part D

Essay Questions

Answer any **1** questions

Each question carries **15** marks

25. Explain the quaternary structure of proteins with suitable examples. [U] / [CO1]

26. Explain the importance of minimum support and confidence thresholds in rule mining. [U] / [CO2]

[15x1 = 15]